

EE 66
Fall 2026

Signals, Dynamics, and Information

Homework 4

This homework is due Friday, September 25, 2026, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

If you do not attach a PDF “printout” of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

1. Euler’s Formula and Trigonometric Identities

The famous Euler’s Formula is defined as follows:

$$e^{it} = \cos(t) + i \sin(t)$$

- (a) Prove Euler’s Formula by starting from the Taylor expansion of e^θ .

Hint: recall the following Taylor expansions:

$$e^\theta = \sum_{n=0}^{\infty} \frac{\theta^n}{n!} = 1 + \theta + \frac{\theta^2}{2!} + \frac{\theta^3}{3!} + \dots$$

$$\cos(t) = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots$$

$$\sin(t) = t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots$$

You will never have to derive these in this class, but hopefully they look familiar.

- (b) Use Euler’s Formula to prove the Double Angle Identity:

$$\cos(2t) = \cos^2(t) - \sin^2(t)$$

- (c) Use Euler’s Formula to prove the Angle Sum Identity:

$$\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$$

- (d) Use Euler’s Formula to prove the hyperbolic cosine function identity, denoted $\cosh(t)$:

$$\cosh(t) = \frac{e^t + e^{-t}}{2}.$$

This pops up all the time in physics and engineering! For example, the shape of a chain hanging under its own weight is modeled with $\cosh$. Further, signal processing engineers sometimes use $\cosh$ when designing filters, and control systems engineers might use $\cosh$ in their loss functions to encourage a system to exhibit some desired behavior. Justify the name of this function.

Hint: recall inverse Euler’s identity, and make a judicious (perhaps imaginary) choice of argument.

2. Roots of Unity and the Fourier Basis

Consider throughout this problem a fixed positive integer p . We define the fundamental frequency

$$\omega_0 = \frac{2\pi}{p},$$

and the p -th root of unity as

$$\omega_p = e^{i\omega_0}.$$

Further, for each integer k , we define the discrete-time complex exponential

$$\psi_k[n] = \omega_p^{kn} = e^{i\omega_0 kn}, \quad n \in \mathbb{Z}.$$

Each ψ_k is p -periodic (prove this!) so we can think of it as a vector in $\mathbb{C}^p$ containing only one period:

$$\psi_k = [\psi_k[0] \quad \psi_k[1] \quad \cdots \quad \psi_k[p-1]]^\top.$$

Convince yourself that this is sufficient to describe ψ_k .

(a) For which integers $k \in \mathbb{Z}$ is $\omega_p^k = 1$? Use your answer to show that

$$\psi_{k+p}[n] = \psi_k[n] \quad \text{for all } n \in \mathbb{Z},$$

so that among $\{\psi_k : k \in \mathbb{Z}\}$ there are only p distinct signals, namely $\psi_0, \psi_1, \dots, \psi_{p-1}$.

(b) Let $m \in \mathbb{Z}$. Show that

$$\sum_{n=0}^{p-1} \omega_p^{mn} = \begin{cases} p & \text{if } m \text{ is a multiple of } p, \\ 0 & \text{otherwise.} \end{cases}$$

Hint: use the geometric-series sum formula.

(c) Show that $\langle \psi_k, \psi_\ell \rangle = 0$ whenever $k \neq \ell$ with $k, \ell \in \{0, 1, \dots, p-1\}$, and compute $\|\psi_k\|^2 = \langle \psi_k, \psi_k \rangle$. We repeat the inner product of complex-valued vectors here for your convenience:

$$\langle u, v \rangle = \sum_{n=0}^{p-1} u[n] v[n]^*.$$

(d) Prove that $\{\psi_0, \psi_1, \dots, \psi_{p-1}\}$ is linearly independent, and conclude that it is a basis for $\mathbb{C}^p$.

Hint: try taking the inner product of $\sum_{k=0}^{p-1} a_k \psi_k = \mathbf{0}$ with an arbitrary ψ_ℓ .

(e) Consider the real-valued signal

$$x[n] = 4 \cos\left(\frac{\pi}{4}n\right) + 2 \sin\left(\frac{\pi}{2}n\right) + 5.$$

For this part only, let $p = 8$. Write x as a linear combination of the signals ψ_k :

$$x[n] = \sum_{k=0}^7 X_k \psi_k[n].$$

In particular, you should clearly identify all eight coefficients $X_0, \dots, X_7$. Then explain, using part (d), why this choice of coefficients is *unique*.

Hint: using part (a), you can write $\psi_{-k} = \psi_{mp-k}$ for any integer m . For example, with $p = 8$, we can choose $m = 1$ to rewrite $\psi_{-1} = \psi_7$.

3. Matching Frequencies

- (a) Let $\omega \in \mathbb{R}$ be a fixed frequency and $\phi_1, \phi_2 \in \mathbb{R}$ be phases. Using Euler's identity, show that

$$\cos(\omega t + \phi_1) + \cos(\omega t + \phi_2)$$

can be written as a single cosine of the same frequency in the form

$$A \cos(\omega t + \phi).$$

and derive expressions for A and ϕ .

Hint: You may find the identity $e^{i\phi_1} + e^{i\phi_2} = e^{i\left(\frac{\phi_1+\phi_2}{2}\right)} \left(e^{i\left(\frac{\phi_1-\phi_2}{2}\right)} + e^{-i\left(\frac{\phi_1-\phi_2}{2}\right)} \right)$ useful.

- (b) In fact, any linear combination of a set of sinewaves of frequency ω is always a sinewave of the same frequency, *even if* each has a distinct phase and amplitude. Prove this. In particular, show that

$$\sum_{k=1}^N A_k \cos(\omega t + \phi_k) = A \cos(\omega t + \phi).$$

Hint: it might be helpful to recognize that

$$A \cos(\omega t + \phi) = \sum_{k=1}^N A_k \cos(\omega t + \phi_k)$$

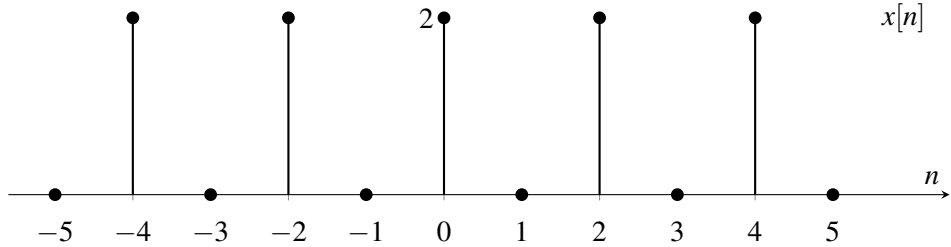
can be obtained by equating the real parts of the two sides of

$$A e^{i(\omega t + \phi)} = \sum_{k=1}^N A_k e^{i(\omega t + \phi_k)}.$$

Note that it is insufficient to assert that the above statement is true; rather, this is equivalent to and perhaps easier-to-prove than your goal. In other words, proving that the complex exponential equality above holds proves that the cosine equality holds.

4. DTFS Warmup (Adapted from EE 120 Discussion)

Consider the periodic DT signal $x[n]$ shown below:



Throughout this problem, we explore the complex exponential discrete-time Fourier series (DTFS) for $x[n]$.

- Determine the fundamental period p and fundamental frequency ω_0 of $x[n]$.
- Determine an expression of the signal $x[n]$ as a linear combination of complex exponentials. Do this by inspection, without resorting to the machinery of the DTFS analysis or synthesis equations.
- Now use the analysis equation to determine the DTFS representation of $x[n]$. That is, determine X_k in the DTFS expansion. For your convenience, the analysis equation is repeated below:

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}. \quad (1)$$

You should use the p and ω_0 found in part (a).

- As a sanity check, use the DTFS coefficients X_k to determine the equation for $x[n]$, and confirm that it matches your answer for part (b). You may find the synthesis equation helpful:

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} \quad (2)$$

Hint: Try splitting up the problem by solving for $x[n]$ when n is even and when it is odd.

5. Signal Analysis with Orthogonal Projections

In many applications, signals are mixtures of a few frequencies. In this problem, you will project a length- p signal onto complex-exponential basis vectors and analyze the resultant coefficients. We define $\omega_0 = \frac{2\pi}{p}$ for the fundamental frequency.

This problem has an IPython component. Please attach a PDF printout of your IPython code to the HW in order to receive credit for problems that involve coding as stated in the submission format above.

(a) Write two Python functions:

- (i) `complex_inner_product(u, v)`: takes two complex vectors $\mathbf{u}, \mathbf{v} \in \mathbb{C}^p$ and returns $\langle \mathbf{u}, \mathbf{v} \rangle$.
- (i) `generate_basis_vector(k, p)`: returns the DTFS basis vector $\boldsymbol{\psi}_k$ of length p with entries $\boldsymbol{\psi}_k[n] = e^{i\omega_0 nk}$.
- (i) Using your functions, with $p = 64$, numerically evaluate $\langle \boldsymbol{\psi}_3, \boldsymbol{\psi}_{10} \rangle$ and report its magnitude.

(b) Let $p = 64$, $f_1 = 2$ Hz, $f_2 = 8$ Hz, $A_1 = 2.0$, $A_2 = 0.5$. Define $t_n = n/p$ and $s: \mathbb{R} \rightarrow \mathbb{R}$ as

$$s(t) = A_1 \cos(2\pi f_1 t) + A_2 \cos(2\pi f_2 t).$$

Write a Python function `generate_signal(p, frequencies, amplitudes)` that returns $\mathbf{s} \in \mathbb{R}^{64}$ with entries $s[n] = s(t_n)$ for these parameters. Additionally report the first four samples $s[0:4]$. Note that instead of s being a discrete-time signal, it is a continuous-time signal for which we take discrete samples, which gives us the vector $\mathbf{s}$.

(c) The contribution of frequency index k in $\mathbf{s}$ is given by the projection coefficient

$$c_k = \text{proj}_{\boldsymbol{\psi}_k}(\mathbf{s}) = \frac{\langle \mathbf{s}, \boldsymbol{\psi}_k \rangle}{\langle \boldsymbol{\psi}_k, \boldsymbol{\psi}_k \rangle} = \frac{1}{p} \langle \mathbf{s}, \boldsymbol{\psi}_k \rangle.$$

Write a Python function `get_projection_coefficient(signal, k, p)` that computes c_k using your tools from part (a).

- (d) For $p = 64$, $k = 0, 1, \dots, 31$, compute c_k and plot $|c_k|$ versus k , with labeled axes. Are there peaks at $k = 2$ and $k = 8$? How do the magnitudes relate to A_1 and A_2 ?
- (e) Repeat parts (b) through (d) with the following frequencies and amplitudes: $f_1 = 24$, $f_2 = 26$, $A_1 = 1$, $A_2 = 1$. We will refer to this signal as s_2 . Afterwards, generate a plot of the discrete-time signal $s_2[n]$. Describe the signal in words. Does the signal match what you'd expect from the DTFS coefficients?

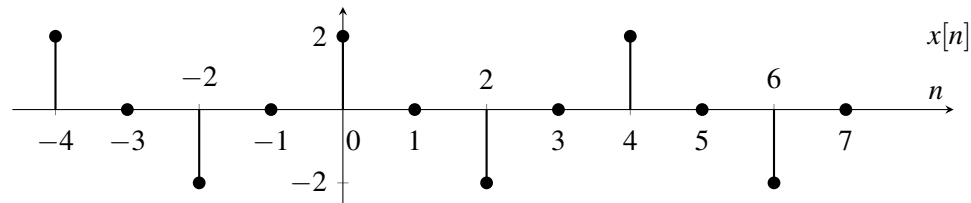
6. DTFS Round Trip

Recall the DTFS analysis and synthesis equations:

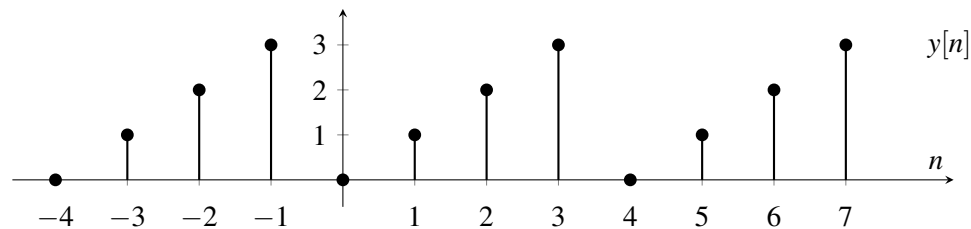
$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n}, X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}.$$

For every subpart, let $p = 4$.

- Determine ω_0 and write out $\psi_0, \psi_1, \psi_2, \psi_3$ explicitly as vectors in $\mathbb{C}^4$.
- Consider the 4-periodic signal $x[n]$ shown below. Compute its DTFS coefficients X_0, X_1, X_2, X_3 .



- Use the synthesis equation first to write $x[n]$ as a sum of complex exponentials. Then simplify your answer to write $x[n]$ in *real* form, i.e., as a constant plus real-valued sinusoids.
- Now let $y[n]$ be the 4-periodic “ramp” shown below. Compute its DTFS coefficients Y_0, Y_1, Y_2, Y_3 .



- As in part (c), write $y[n]$ as a sum of complex exponentials and then in real form.
Hint: You should have noticed that $Y_3 = Y_1^$ (if not, go back and check your work!). Try to express Y_1 as a complex exponential, i.e., write $Y_1 = |Y_1| e^{i\angle Y_1}$ and $Y_3 = Y_1^* = |Y_1| e^{-i\angle Y_1}$. These facts will help you combine the $k = 1$ and $k = 3$ terms to create a single cosine.*
- For each of the two signals above, sketch a stem plot of $|X_k|$ (respectively, $|Y_k|$) versus k . Notice that some of the X_k s are 0, whereas all of the Y_k s are non-zero (if not, go back and check your work!). Using the original signals, justify this observation.

In particular, you should compare characteristics of the original signal which explain the fact that $y[n]$'s DTFS expansion has a constant term whereas $x[n]$'s does not. Make a similar argument to explain why $y[n]$'s DTFS expansion needs the fastest-varying basis signal $(-1)^n$ but $x[n]$ does not.

Hint: try plotting the signal you'd get by removing the ψ_2 term from $y[n]$'s DTFS expansion. What does this look like? In general, what contribution does a rapidly oscillating term like $(-1)^n$ make to a signal's appearance?

7. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

- (a) We just finished Exam 1. Congrats! We would love to hear some feedback while the experience is “fresh.” Perhaps you already had a chance to share your thoughts with course staff during discussion or office hours. Regardless (in case you did not, or if they forgot what you shared) we are hoping you will write something down here. We will do what we can to make the next exam better!
- (b) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*
- (c) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.
- (d) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?
- (e) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able to practice in this assignment?
- (f) Is there anything else you would like to share?